

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (CSSP)	Indian Institute of Technology, Patna	7.69	2024 - Present
B.Tech. (ECE)	MIT World Peace University, Pune	9.48	2017 - 2021

EXPERIENCE

Tata Consultancy Services
Assistant System Engineer Trainee

Feb 2022 - July 2023
Kolkata, India

- Optimized SQL based data pipelines to improve data availability for analytics and reporting workflows.
- Developed automation tools (Java) to streamline reporting processes, reducing manual effort by 20%.
- Performed data-driven root cause analysis on production systems to identify performance issues and improve reliability.
- Collaborated in Agile teams to support maintenance and delivery of large-scale production systems.
- Worked with structured datasets and system logs to support analytical workflows and troubleshooting.

InfoVision Labs India Pvt. Ltd.
Software Engineer LI, FTE

Sept 2021
Pune, India

- Wrote test cases and debugged Python modules to improve code quality and reliability.
- Gained practical exposure to software testing lifecycle (STLC).

PROJECTS

Hybrid Neuro-Symbolic Vehicle Anomaly Detection Platform ([Link](#))
IIT Patna

Dec 2025 - Present
IIT Patna

- Built a vehicle behavioral threat classification system (Normal / Distracted / Aggressive) from aerial footage using fused ViT visual features and physics-based kinematic divergence scoring.
- Implemented a RAG pipeline using FAISS VectorDB and SQLite for contextual anomaly memory retrieval, historical event search, and incident analysis across sequential video inference.
- Engineered a tactical surveillance dashboard with live vehicle lock-on, 4× zoom tracking, bearing/range readouts, dual-signal anomaly charts, and real-time threat profiling.
- Implemented Gemini API-based agentic workflows to generate contextual tactical responses from fused visual and motion signals.

Autonomous Multi-Agent AI for End-to-End Marketing Decisions ([Github](#))
IIT Patna

Dec 2025
IIT Patna

- Architected a self-evaluating multi-agent AI platform using Gemini-based LLM orchestration to transform raw marketing data into KPI analysis, forecasting, strategic actions, and risk assessment.
- Built coordinated AI agents for analysis, forecasting, planning, evaluation, and recommendation refinement; implemented structured JSON-constrained generation with response validation and fault-tolerance.
- Designed self-refinement and evaluation-feedback pipelines improving reasoning quality across iterative cycles.

AeroViT: Transformer-CNN Framework for Aerial Anomaly Detection
IIT Patna

July 2025 - Dec 2025
IIT Patna

- Developed a hybrid Transformer-CNN anomaly detection framework achieving 89% AUC and approximately 58× improvement in computational efficiency.
- Designed hybrid perceptual-loss pipelines combining reconstruction consistency, SSIM, and feature-level anomaly localization.
- Implemented FFT-based temporal evaluation, motion-consistency analysis, and embedding-space diagnostics.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, JavaScript, HTML/CSS, Java

AI & LLM Engineering: LLMs & Embeddings, RAG Pipelines, Prompt Engineering, Structured Outputs, Multi-Agent AI Systems, Agentic Workflows

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, Ultralytics YOLO, Vision Transformers (ViT)

Vector & Data Systems: FAISS VectorDB, SQLite, Pandas, NumPy, Statistical Analysis, Data Visualization

Domains: AI-Native Engineering, Computer Vision, Anomaly Detection, Real-Time Analytics, Enterprise AI Systems

ACHIEVEMENTS

- Qualified GATE 2024